

Data science is an [interdisciplinary](#) academic field^[1] that uses [statistics](#), [scientific computing](#), [scientific methods](#), processing, [scientific visualization](#), [algorithms](#), and systems to extract or extrapolate [knowledge](#) from potentially noisy, [structured](#), or [unstructured data](#).^[2]

Data science also integrates domain knowledge from the underlying application domain (e.g., natural sciences, information technology, and medicine).^[3] Data science is multifaceted and can be described as a science, a research paradigm, a research method, a discipline, a workflow, and a profession.^[4]

Data science is "a concept to unify [statistics](#), [data analysis](#), [informatics](#), and their related [methods](#)" to "understand and analyze actual [phenomena](#)" with [data](#).^[5] It uses techniques and theories drawn from many fields within the context of [mathematics](#), statistics, [computer science](#), [information science](#), and [domain knowledge](#).^[6] However, data science is distinct from [computer science](#) and information science. [Turing Award](#) winner [Jim Gray](#) imagined data science as a "fourth paradigm" of science ([empirical](#), [theoretical](#), [computational](#), and now data-driven) and asserted that "everything about science is changing because of the impact of [information technology](#)" and the [data deluge](#).^{[7][8]}

Data science is often described as a multidisciplinary and rapidly evolving field because it draws on techniques from diverse areas, such as computer science, statistics, information science, and other subject-specific disciplines. Some researchers argue that the combination of the different fields resembles how information science was decades ago (Mayernik, 2023). These similarities help clarify how data science gradually became its own field of study.^[9]

A **data scientist** is a professional who creates programming code and combines it with statistical knowledge to summarize data.^[10]

Foundations

Data science is an [interdisciplinary field](#)^[11] focused on [extracting knowledge](#) from typically [large data sets](#) and applying the knowledge from that data to solve problems in other application domains. The field encompasses preparing data for analysis, formulating data science problems, [analyzing](#) data, and summarizing these findings. As such, it incorporates skills from computer science, mathematics, [data visualization](#), [graphic design](#), [communication](#), and [business](#).^[12]

[Vasant Dhar](#) writes that statistics emphasizes quantitative data and description. In contrast, data science deals with quantitative and qualitative data (e.g., from images, text, sensors, transactions, customer information, etc.) and emphasizes prediction and action.^[13] [Andrew Gelman](#) of [Columbia University](#) has described statistics as a non-essential part of data science.^[14] Stanford professor [David Donoho](#) writes that data science is not distinguished from statistics by the size of datasets or use of computing and that many graduate programs

misleadingly advertise their analytics and statistics training as the essence of a data-science program. He describes data science as an applied field growing out of traditional statistics.^[15]

Etymology

Early usage

In 1962, [John Tukey](#) described a field he called "[data analysis](#)", which resembles modern data science.^[15] In 1985, in a lecture given to the Chinese Academy of Sciences in Beijing, [C. F. Jeff Wu](#) used the term "data science" for the first time as an alternative name for statistics.^[16] Later, attendees at a 1992 statistics symposium at the [University of Montpellier II](#) acknowledged the emergence of a new discipline focused on data of various origins and forms, combining established concepts and principles of statistics and data analysis with computing.^{[17][18]}

The term "data science" has been traced back to 1974, when [Peter Naur](#) proposed it as an alternative name to computer science. In his 1974 book *Concise Survey of Computer Methods*, Peter Naur proposed using the term 'data science' rather than 'computer science' to reflect the growing emphasis on data-driven methods.^{[19][6]} In 1996, the International Federation of Classification Societies became the first conference to specifically feature data science as a topic.^[6] However, the definition was still in flux. After the 1985 lecture at the Chinese Academy of Sciences in Beijing, in 1997 [C. F. Jeff Wu](#) again suggested that statistics should be renamed data science. He reasoned that a new name would help statistics shed inaccurate stereotypes, such as being synonymous with accounting or limited to describing data.^[20] In 1998, Hayashi Chikio argued for data science as a new, interdisciplinary concept, with three aspects: data design, collection, and analysis.^[18]

Modern usage

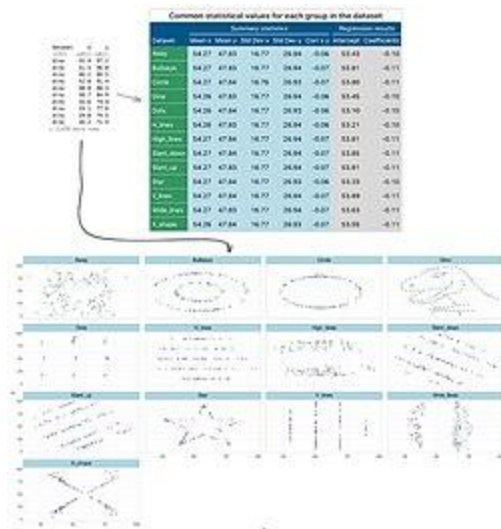
In 2012, technologists [Thomas H. Davenport](#) and [DJ Patil](#) declared "Data Scientist: The Sexiest Job of the 21st Century",^[21] a catchphrase that was picked up even by major-city newspapers like the [New York Times](#)^[22] and the [Boston Globe](#).^[23] A decade later, they reaffirmed it, stating that "the job is more in demand than ever with employers".^[24]

The modern conception of data science as an independent discipline is sometimes attributed to [William S. Cleveland](#).^[25] In 2014, the [American Statistical Association](#)'s Section on Statistical Learning and Data Mining changed its name to the Section on Statistical Learning and Data Science, reflecting the ascendant popularity of data science.^[26]

Over the last few years, many colleges have begun to create more structured undergraduate programs in data science. According to a report by the National Academies, strong programs typically include training in statistics, computing, ethics, and communication, as well as hands-on work in a specific field (National Academies of Sciences, Engineering, and Medicine, 2018). As schools try to prepare students for jobs that use data, these practices become more common.^[27]

The professional title of "data scientist" has been attributed to [DJ Patil](#) and [Jeff Hammerbacher](#) in 2008.^[28] Though it was used by the [National Science Board](#) in their 2005 report "Long-Lived Digital Data Collections: Enabling Research and Education in the 21st Century", it referred broadly to any key role in managing a digital [data collection](#).^[29]

Data science and data analysis



Example of [exploratory data analysis](#) using the [Datasaurus dozen](#) data set

In data science, **data analysis** is the process of inspecting, cleaning, transforming, and modelling data to discover useful information, draw conclusions, and support decision-making.^[30] It includes **exploratory data analysis** (EDA), which uses graphics and descriptive statistics to explore patterns and generate hypotheses,^[31] and **confirmatory data analysis**, which applies statistical inference to test hypotheses and quantify uncertainty.^[32]

Typical activities comprise:

- data collection and integration;
- data cleaning and preparation (handling missing values, outliers, encoding, normalisation);
- feature engineering and selection;
- visualisation and descriptive statistics;^[31]
- fitting and evaluating statistical or machine-learning models;^[32]
- communicating results and ensuring reproducibility (e.g., reports, notebooks, and dashboards).^[33]

Lifecycle frameworks such as [CRISP-DM](#) describe these steps from business understanding through deployment and monitoring.^[34]

Data science involves working with larger datasets that often require advanced computational and statistical methods to analyze. Data scientists often work with [unstructured data](#) such as text or images and use [machine learning](#) algorithms to build predictive models. Data science often uses [statistical analysis](#), [data preprocessing](#), and [supervised learning](#).^{[35][36]}

Recent studies indicate that AI is moving towards data-centric approaches, focusing on the quality of datasets rather than just improving AI models. This trend focuses on improving system performance by cleaning, refining, and labeling data (Bhatt et al., 2024). As AI systems grow larger, the data-centric view has become increasingly important.^[37]